CODECHECK -- Independent execution of computations underlying research articles

Stephen J Eglen
https://sje30.github.io
sje30@cam.ac.uk

Cambridge Computational Biology Institute
University of Cambridge
@StephenEglen

Slides: http://bit.ly/eglen-ssi (CC-BY license)

Acknowledgements

Joint work with Daniel Nüst; Supported by Mozilla foundation and SSI https://codecheck.org.uk.

One slide overview.

1. Author sends codechecker a link to their (open access) paper, data and code, along with instructions on how to run code.

2. Codechecker runs code independently, and checks if key results (figures, tables) can be reproduced in own environment. If no, go to step 3, else

Progress to date

1. Summer 2019: two students helped curate some "historical" papers of interest in my field (computational neuroscience).

2. April 2020: publication of first real certificate, in Gigascience for a machine learning paper blog

3. COVID-19: started working on reproducing key papers, firstly from LSHTM and then Imperial College.

Sabine L.van Elsland
@SabineLvE

Independent review @StephenEglen confirmed that @MRC_Outbreak team's #COVID19 simulation is reproducible: thumbs up from code-checking efforts @nature #COVID19 #covid19sciencenature.com/articles/d4158…

Next steps

1. Writing a paper.
2. Discuss with journals about embedding this practice into publication workflow.
3. Discuss with funders about extending project.
4. Building community of codecheckers.